APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SIXTH SEMESTER B.TECH DEGREE (R,S) EXAMINATION, July 2023(2019 Scheme)

Course Code: HUT310

Course Name: MANAGEMENT FOR ENGINEERS

Max. Marks: 100		Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. Marks

1 What are the characteristics of Management? (3)

- Goal-oriented.
- Pervasive.
- Multi-dimensional.
- Continuous process.
- Group activity.
- Dynamic function.
- Intangible force

2 Explain Contingency Approach in Management (3)

Contingency management is based on the idea that there is no one-size-fits-all approach to management. It involves developing alternative plans that can be implemented if unexpected circumstances arise, rather than relying on one fixed plan of action.It is also called “if then” approach

3 Differentiate between Manager and Leader (3)

A manager tends to focus on controlling resources and optimising processes, while a leader focuses on inspiring and empowering people to work together towards a common goal.

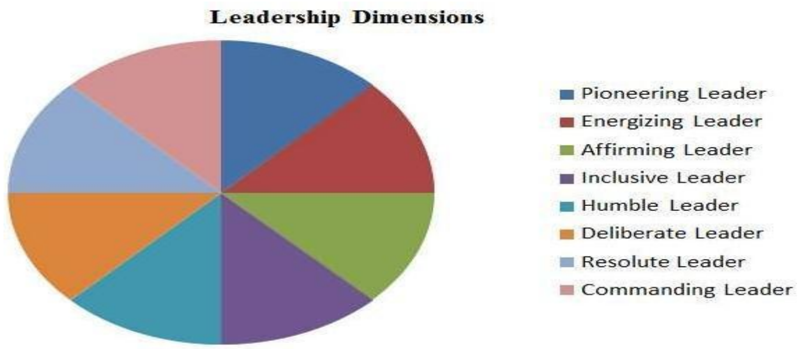
Leaders	Managers
Leaders will focus on objectives which are developed by managers.	Managers will develop business objectives for the organization.
A leader interacts with the employees.	A manager interacts with the leader.
A leader works with team.	A manager gives instructions to groups.
Leaders implement the ideas of the managers.	Managers invent the ideas.
They focus on employees	They focus on resources.
Their main responsibility to inspire employees	Their main function to develop plans

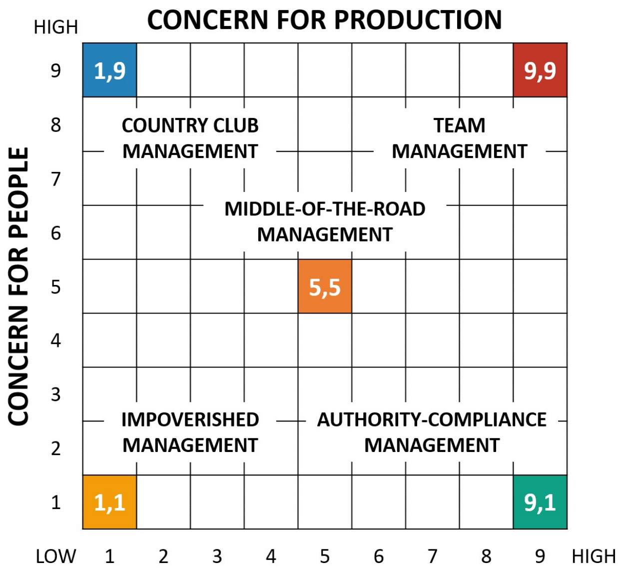
4		What are the different types of plans? **<u>Based on the frequency of use</u>** – Single-use plans – Standing plans **<u>Based on scope</u>** – Operational Planning – Tactical Planning – Strategic Planning **<u>Based on time frame</u>** – Long-term plans – Intermediate plans – Short-term plans	(3)
5		Productivity is commonly defined as a ratio between the output volume and the volume of inputs. In other words, it measures how efficiently production inputs, such as labour and capital, are being used in an economy to produce a given level of output. Objectives 1.To ensure resource efficiency 2. To analyze the growth of the firm 3,To take up corrective actions 4.As a means of communicating the growth and development to stakeholders	(3)

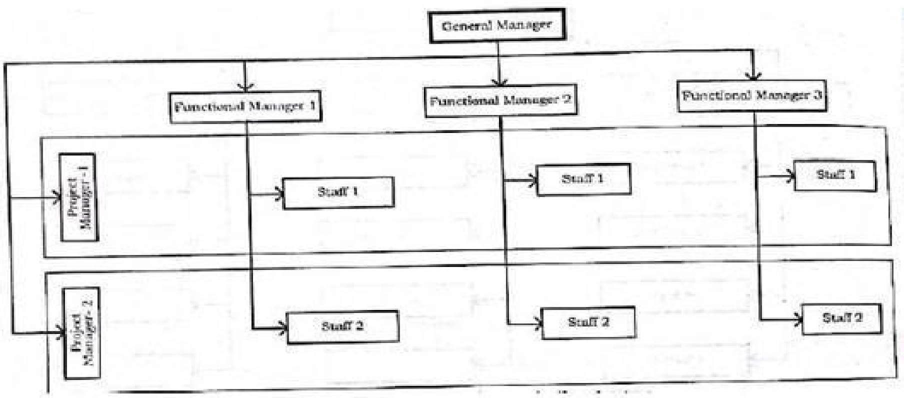
6		What are the factors affecting Decision Making? 1.Decision making 2.Uncertainity about future 3.Time of decision making	(3)
7		Differentiate between critical and non-critical activity ***Critical activities*** – In a network diagram, critical activities are those, if consume more than their estimated time, the project will be delayed. (1.5 marks) ***Non-critical activities*** – Such activities have provision (float or slack), so that even if they consume time over and above the estimated time, the project will not be delayed. (1.5 marks)	(3)
8		List the steps involved in PERT method. ***Steps involved in PERT*** 1. The project is broken down into various activities systematically. 2. Arrange all activities in logical sequence. 3. Construct the network diagram. 4. Events and activities are numbered. 5. Using three time estimate, the expected time for each activity is calculated. 6. Standard Deviation and variance of each activity are calculated. 7. ***EST*** and ***LFT*** are calculated. 8. Expected time, earliest starting time and latest finishing times are marked on the network diagram. 9. Slack is calculated. 10. Critical paths are identified and marked on the network diagram. 11. Length of critical path or total project duration is found out. 12. Lastly the probability that the project will finish at due date is calculated	(3)
9		Write a short note on Corporate Social Responsibility Corporate social responsibility (CSR) refers to strategies that companies put into action as part of corporate governance that are designed to ensure the company's operations are ethical and beneficial for society.	(3)

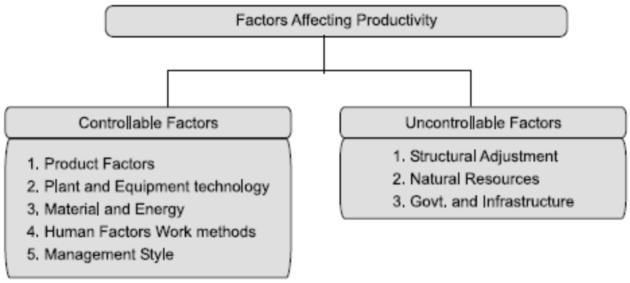
10		Distinguish between Marketing and Sales ***Marketing activity*** finds out the needs of customers so as to deliver the company's products to the customers more effectively and efficiently than its competitors i.e., any activity which facilitates selling is called marketing. (1.5 marks) ***Selling activity*** is concerned with the sales of the company's product or selling activity concentrates on pushing the products to the customer whereby focuses on the needs(targets)set by the seller. (1.5 marks)	(3)

PART B			
Answer one question from each module, each carries 14 marks.			
		Module I	
11		1. Division of work 2. Authority and Responsibility 3. Discipline 4. Unity of Command 5. Unity of Direction 6. Subordination of Individual Interest to General Interest 7. Remuneration 8. Centralization 9. Scalar Chain 10. Order 11. Equity 12. Stability of tenure of personnel 13. . Initiative 14. Esprit de Corps **Any 10 points- 14 marks**	(14)
		OR	
12	a)	Roles of a manager Interpersonal Role Informational Role Decision Role –	(7)
	b)	Skills required for a manager ● CONCEPTUAL SKILL ● TECHNICAL SKILL ● HUMAN RELATION SKILL	(7)
		Module II	

13	a)	Dimensions of leadership 8 dimension * 1 mark = 8 marks 	(8)

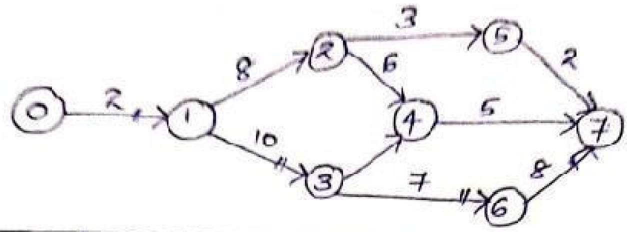
	b)	Explanation- 4 marks, figure- 2 marks 	(6)
		OR	
14	a)	Steps in Organizing 1.Identify the work to be performed 2.Classify the work or group the work 3.Assigning activities to the group or individuals 4.Deciding the hierarchy of authority	(8)

	b)	**Matrix Organization**  Explanation- 4 marks, figure- 2 marks	(6)

		Module III	
15	a)	Explain various factors that affect Productivity.. 1. Controllable (or internal) factors 2. Un-controllable (or external) factors. 	(8)
	b)	A health club has two employees who work on lead generation. Each employee works 40 hours a week, and is paid $20 an hour. Each employee identifies an average of 400 possible leads a week from a list of 8,000 names. Approximately 10 percent of the leads become members and pay a one-time fee of $100. Material costs are $130 per week, and overhead costs are $1,000 per week. Calculate the multifactor productivity for this operation in fees generated per dollar of inputMFP=Quantity Ouput/ (Labor cost+ Material cost + Overhead cost) Quantity Ouput = (Possible leads)* (No. of workers Fee)*(fee)*(conversion percentage) = (400*2*$100*0.1) = $8000 (2 marks) Labor cost = (2*40*$20) Material cost = $130 Overhead cost= $1000 Labor cost+ Material cost + Overhead cost = (2*40*$20)+$130+$1000=$2730 (2 marks) MFP= $8000/$2730 = 2.93 (2 marks)	(6)
		OR	
16	a)	Decision Making under certainty (3 marks) A condition of certainty exists when the decision-maker knows with reasonable certainty what the alternatives are, what conditions are associated with each alternative, and the outcome of each alternative. Under conditions of certainty, accurate, measurable, and reliable information on which to base decisions is available.	(9)

		Decision Making under Risk (3 marks) When a manager lacks perfect information or whenever an information asymmetry exists, risk arises. Under a state of risk, the decision maker has incomplete information about available alternatives but has a good idea of the probability of outcomes for each alternative. Decision Making under uncertainty (3 marks) Most significant decisions made in today's complex environment are formulated under a state of uncertainty. Conditions of uncertainty exist when the future environment is unpredictable and everything is in a state of flux. The decision-maker is not aware of all available alternatives, the risks associated with each, and the consequences of each alternative or their probabilities.	
	b)	Explain the importance of decision making in Management. 1. Better Utilisation of Resources 2. Facing Problems and Challenges 3. Business Growth 4. Achieving Objectives 5. Increases Efficiency 6. Facilitate Innovation 7. Motivates Employee	(5)
17	a)	**Module IV** Draw a network diagram for the following project given below. Find the critical path and calculate the project duration & float of the network.	

17 a)



Activity	Duration	EST	LFT	LST	EFT	Float
0-1	2	0	2	0	2	0
1-2	8	2	16	8	10	6
1-3	10	2	12	2	12	0
2-4	6	10	22	16	16	6
3-4	3	12	22	19	15	7
2-5	3	10	25	22	13	12
3-6	7	12	19	12	19	0
5-7	2	13	27	25	15	12
6-7	8	19	27	19	27	0
4-7	5	16	27	22	21	6

Critical Path : 0-1-3-6-7

Project Duration 27 days

Network diagram-3 marks, Table- 3 marks, critical path- 1 mark, project duration- 1 mark

b) Compare CPM and PERT method of Evaluation.

Abbreviation	
PERT – Project Evaluation and Review Technique	CPM – Critical Path Method
What does It Mean?	
PERT – PERT is a popular project management technique that is applicable when the time required to finish a project is not certain	CPM – CPM is a statistical algorithm which has a certain start and end time for a project
Model Type	
PERT – PERT is a probabilistic model	CPM – CPM is a deterministic model
Focus	
PERT – The main focus of PERT is to minimise the time required for completion of the project	CPM – The main focus of CPM is on a trade-off between cost and time, with a major emphasis on cost-cutting.
Orientation type	
PERT – PERT is an event-oriented technique	CPM – CPM is an activity-oriented technique

OR

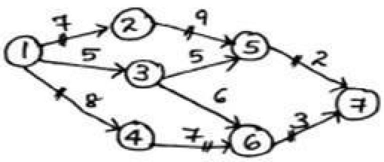
18 a) Consider the data of a project as shown in table. If the indirect cost per week is Rs.200, Find the optimal crashed project completion time.

18 a) (8)

Soln: Slope = $\dfrac{\text{Crash cost} - \text{Normal cost}}{\text{Normal Time} - \text{Crash Time}}$

Activity	Slope
1-2	50
1-3	100
1-4	200
2-5	225
3-5	150
3-6	200
4-6	125
5-7	100
6-7	350

I iteration:



Critical path

1-2-5-7 ⇒ 7+9+2 = 18

1-3-5-7 ⇒ 5+5+2 = 12

1-3-6-7 ⇒ 5+6+3 = 14

1-4-6-7 ⇒ 8+7+3 = 18

∴ Critical Paths are 1-2-5-7 & 1-4-6-7

Normal Project Completion time = 18 weeks

Critical paths : 1-2-5-7 & 1-4-6-7

Total Cost = Σ normal cost + (18 × indirect cost)

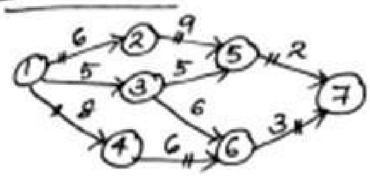
= Rs. 6,500 + (18 × 200) = Rs. 10,100

Crash limit & Slope

Critical Path	Critical activity	Crash limit = Crash time − Normal time	Cost slope	
1-2-5-7	1-2	3	50	least = 50
	2-5	2	225	
	5-7	1	100	
1-4-6-7	1-4	3	200	least = 125
	4-6	2	125	
	6-7	1	350	

Crash 1-2 from 1-2-5-7 by 1 week & 4-6 from 1-4-6-7 1 week

II iteration :



Critical path

1-2-5-7 ⟹ 6+9+2 = 17

1-3-5-7 ⟹ 5+5+2 = 12

1-3-6-7 ⟹ 5+6+3 = 14

1-4-6-7 ⟹ 8+6+3 = 17

∴ critical paths are 1-2-5-7 & 1-4-6-7

Normal Project completion time = 17 weeks

critical paths : 1-2-5-7 & 1-4-6-7

Total cost = Previous cost + direct cost (slope cost) – indirect cost

= Rs. 1,100 + (50+125) – 200 = Rs. 10,075

Crash limit & slope

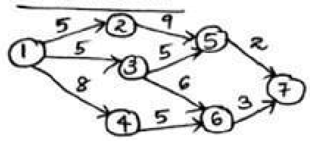
critical path	critical activity	crash limit	cost slope	
1-2-5-7	1-2	2	50	least cost = 50
	2-5	2	225	
	5-7	1	100	
1-4-6-7	1-4	3	200	least cost = 125
	4-6	1	125	
	6-7	1	350	

crash 1-2 from 1-2-5-7 by 1week & 4-6 from 1-4-6-7 by 1week

Normal time of 1-2 = 6-1 = 5week

ie 6-1 = 5 weeks

III iteration



Critical path

1-2-5-7 ⇒ 5+9+2 = 16

1-3-5-7 ⇒ 5+5+2 = 12

1-3-6-7 ⇒ 5+6+3 = 14

1-4-6-7 ⇒ 8+5+3 = 16

Normal Project completion time = 16 weeks

∴ critical paths are 1-2-5-7 and 1-4-6-7

critical paths : 1-2-5-7 & 1-4-6-7

Total cost = Previous cost + (50+125) — indirect cost

= Rs.10,075 + Rs.175 − Rs.200 = Rs.10,050

Crash limit & slope

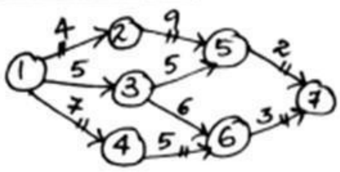
Critical path	Critical activity	Crash limit	Cost slope	
1-2-5-7	1-2	1	50	least = 50
	2-5	2	225	
	5-7	1	100	
1-4-6-7	1-4	3	200	Next least = 200
	4-6	0	125	
	6-7	1	350	

Crash 1-2 by 1week

Normal week = 15 week

Crash 1-4 by 1week

Normal week = 15 week

		IV iteration  Critical path 1-2-5-7 ⇒ 4+9+2 = 15 1-3-5-7 ⇒ 5+5+2 = 12 1-3-6-7 ⇒ 5+6+3 = 14 1-4-6-7 ⇒ 7+5+3 = 15 ∴ Critical paths are 1-2-5-7 and 1-4-6-7 Normal project completion time = 15 weeks Critical paths : 1-2-5-7 & 1-4-6-7 Total cost = Rs.10,050 + (50+200) - 200 = Rs.10,100 Total cost becomes greater than previous one. ∴ Project cost = Rs. 10,050 Final crashed project completion time = 16weeks **Each iteration carries 2 marks each**	
	b)	Explain the following terms (i) Crash Cost (ii) Crash time (iii) Project Crashing ***Crash time*** – The crash time refers to the shortest possible time to complete an activity with additional resources. (2 marks) ***Crash cost*** – The crash cost refers to the activity cost under the crashing activity time. (2 marks) The process of reducing the time of the project by reducing the scheduled time of some of the activities is called ***crashing.*** (2 marks)	(6)
		Module V	
19	a)	What are the objectives of Operation Customer Service Resource utilization 1. Customer service. Customers demand better quality, greater speed, and lower costs satisfy the customer in terms of cost and timing. Provide ‘ right thing at right price at the right time. 2. Resource utilization: it measures performance and effort over an amount of	(4)

		available time (or capacity). Optimal resource utilization allows project managers to foresee resource availability across multiple categories 	
	b)	Explain the scope of Operation Management Production and planning control Quality control Materials management Maintenance management	(10)

		OR	
20	a)	Explain different types of entrepreneurs 1) According to types of business: Business entrepreneurs. Trading entrepreneurs. Industrial entrepreneurs. Corporate entrepreneurs. Agricultural entrepreneurs. 1) According to use of technology: Technical entrepreneurs. Professional entrepreneurs. Non-technical entrepreneurs. High tech entrepreneurs. 2) According to motivation of entrepreneurs: Pure entrepreneurs. Induced entrepreneurs. Motivated entrepreneurs. Spontaneous entrepreneurs. 3)According to growth: Growth entrepreneurs. Super growth entrepreneurs. 4) According to stages of development: 1st generation entrepreneurs. Modern entrepreneurs. Classical entrepreneurs. 5) According to scale of operation: Small scale entrepreneurs. Large scale entrepreneurs.	(7)

	b)	What are the characteristics of a successful Entrepreneur High degree of commitment. High energy level. Foresightedness. Desire for responsibility. Risk taking ability. Leadership and managerial skills. Value for achievement over money. Open-mindedness and optimism.	(7)

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